



Name: Cohort/Batch: Date: Score:

GITHUB ACTIONS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A DevOps

TOTAL: 10 QUESTIONS

Q1 What problem does GitHub Actions solve in DevOps or Infrastructure? **Great Model**

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Q6 How does GitHub Actions integrate with CI/CD pipelines? **Availability**

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Q2 What is GitHub Actions's core architecture? **Architecture**

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Q7 How do you debug a failed GitHub Actions deployment? **Lifecycle**

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Q3 How does GitHub Actions define infrastructure or configuration as code? **Configuration**

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Q8 What are security best practices for GitHub Actions? **Compliance**

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Q4 How does GitHub Actions ensure idempotency? **Hardening**

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Q9 How does GitHub Actions scale for large environments? **Testing**

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Q5 How do you handle secrets in GitHub Actions? **SSO / IdP**

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Q10 What are common mistakes teams make when adopting GitHub Actions? **Tuning**

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 GitHub Actions addresses a specific concern

Mitigation

GitHub Actions addresses a specific concern provisioning, configuration management, orchestration, or CI/CD by automating manual steps that are error-prone, slow, or not reproducible when done by hand.

A6 CI/CD pipelines call GitHub Actions in automated

Observability

CI/CD pipelines call GitHub Actions in automated steps: plan on a pull request to preview changes and apply on merge to main. Approval gates can require a human to review the plan before changes go to production.

A2 GitHub Actions has a control plane that

Primitives

GitHub Actions has a control plane that stores desired state and a data plane (agents or push-based SSH) that enforces it. Understanding this distinction clarifies where configuration is evaluated and where changes are applied.

A7 Check GitHub Actions's logs for the specific

Automation

Check GitHub Actions's logs for the specific error message and the resource that failed. For infrastructure tools, re-run with increased verbosity (-v, --debug). Review the state file or event history to understand what changed before the failure.

A3 GitHub Actions uses declarative files (YAML, HCL,

Hardening

GitHub Actions uses declarative files (YAML, HCL, or a DSL) to express desired state. A plan/diff phase shows what will change before applying. Version-controlling these files enables peer review and rollback.

A8 Rotate credentials used by GitHub Actions, restrict

Governance

Rotate credentials used by GitHub Actions, restrict network access to its control plane, enable audit logging of all operations, and use least-privilege service accounts. Never run GitHub Actions with admin credentials in production.

A4 GitHub Actions operations are designed to produce

Gotchas

GitHub Actions operations are designed to produce the same result when run multiple times. Applying the same config twice converges to the desired state rather than creating duplicate resources. This makes automation safe to re-run.

A9 Large GitHub Actions deployments use workspaces

Assessment

Large GitHub Actions deployments use workspaces or namespaces to isolate environments, remote state backends for team collaboration, and modular code to avoid a single monolithic config that is slow to plan/apply.

A5 Secrets should never be stored in plain

Federation

Secrets should never be stored in plain text in GitHub Actions config files. Use a secrets backend (Vault, AWS Secrets Manager) and inject values at runtime via environment variables or encrypted vaults supported by the tool.

A10 Common pitfalls include storing secrets in plain

Optimization

Common pitfalls include storing secrets in plain text config, skipping the plan step before apply, not reviewing state drift, and not having a rollback strategy when a deployment fails partway through.